

Aditya Kulkarni

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SUMMARY

Backend engineer with production TypeScript/Python experience across LLM evaluation, real-time ingestion pipelines, and observability. Built a vendor-neutral LLM eval harness tracking hallucination rate, p50/p95 latency, and cost-per-call across Claude, GPT-4o, and Gemini with paired bootstrap significance testing. Kafka-driven ingestion pipelines (VectorLift); Prometheus/Grafana/OpenTelemetry observability (PulseRank); 2K async LLM req/sec (AlgoPulse). Shipped 10+ features at Isomer AI (394 commits, 7 deployments, TypeScript/Node.js). MS CS NC State (May 2026).

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Shipped 10+ backend features across 7 production deployments in TypeScript, Node.js, Express, PostgreSQL, Redis, and Docker; built microservices with CI/CD via GitHub Actions, Sentry observability, and real-time event pipelines (Pusher) -- 394 commits on a founding-stage team.
 - Implemented Temporal.io durable workflow orchestration, multi-model LLM routing (Claude/GPT-4o/Gemini, 8+ models) via MCP, Auth0 RBAC, and document processing pipelines (PDF generation, email parsing, S3 storage).
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Designed and deployed a cloud-native genomic data platform on Google Cloud Run serving 500GB of datasets to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.
- Software Developer Intern

Mphasis (HP Company)

Jun 2024 - Aug 2024

 - Built AI-driven automation workflows including an ML-powered health claim processing tool (+25% accuracy); automated CI/CD on AWS and GCP, integrated Grafana monitoring, and supported model validation and red-teaming.
- AI/ML Intern

Capgemini

Jun 2023 - Aug 2023

 - Built distributed data ingestion pipelines using Hadoop and Apache Spark for large-scale batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and latency by 20%.

PROJECTS

- LLM Eval Harness

Evals / Observability

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000 frozen examples (grounded QA, multi-hop QA, FEVER claim verification). Tracked hallucination rate, p50/p95 latency, and cost-per-call. Paired bootstrap CIs, McNemar test, Holm correction for valid model comparison. DuckDB + Parquet + Streamlit regression dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker
- VectorLift -- Ingestion Pipeline

Pipelines / Observability

Kafka-driven real-time corpus ingestion into Elasticsearch and Qdrant; BM25 baseline, dense bi-encoder retrieval, and cross-encoder re-ranker; FastAPI serving with per-stage latency tracing; Prometheus + Grafana observability; NDCG@10/MRR@10 evaluation across 6 pipeline configurations.

Python, PyTorch, Kafka, Elasticsearch, Qdrant, FastAPI, Prometheus, Grafana, Docker
- PulseRank -- Real-Time Platform

Distributed / High-Availability

CQRS event-driven platform: MySQL outbox + idempotency keys for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-ms reads; SSE live push; k6 benchmarks (10K users); OpenTelemetry tracing, Prometheus + Grafana observability.

TypeScript, NestJS, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, Grafana
- AlgoPulse -- High-Throughput Orchestration

Real-Time / Scale

Re-architected from API polling to event-driven streaming after diagnosing throughput bottleneck; Temporal.io durable workflow state with fault-tolerant retries, activity checkpointing, and saga-pattern rollback; Kafka for partitioned worker coordination. Reached 2,000 async LLM inference req/sec with zero message loss.

TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka, Docker

EDUCATION

- North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms
- Mumbai University -- BTech Computer Science | CGPA: 9.32/10

Aug 2020 - May 2024

Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain

SKILLS & AWARDS

Languages: TypeScript, Python, JavaScript, Java, SQL, C | **Backend:** Node.js, Express, FastAPI, NestJS | **Pipelines / Infra:** Kafka, Spark, Temporal.io, Docker, AWS, GCP, CI/CD | **Observability:** Prometheus, Grafana, OpenTelemetry, Sentry | **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant | **AI/ML:** LLMs, MCP, RAG, LangChain, PyTorch, OpenAI/Anthropic/Google SDKs

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)